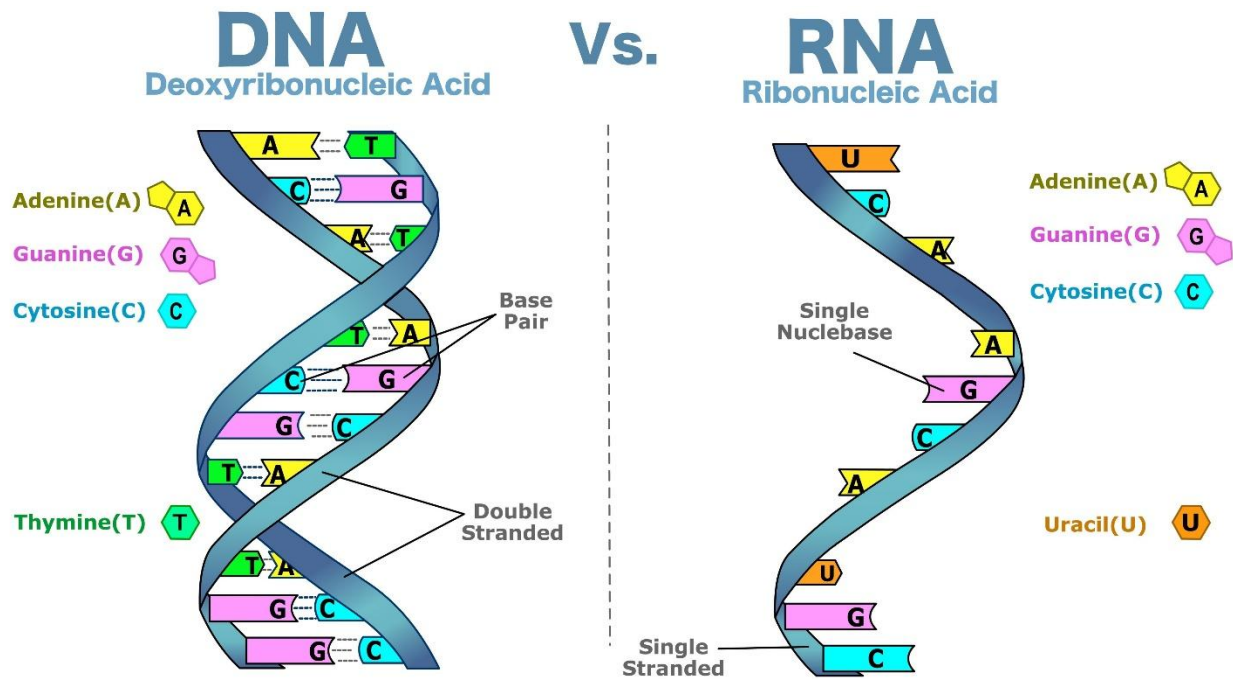


BIOINFORMATICS

WEEK 2 - DAY 16



Characteristic	DNA (Deoxyribonucleic Acid)	RNA (Ribonucleic Acid)
Primary Function	Long-term storage of genetic blueprints.	Direct execution of protein synthesis.
Strand Structure	Double-stranded double helix.	Usually single-stranded ribbon.
Pentose Sugar	Deoxyribose sugar.	Ribose sugar.
Nitrogenous Bases	Adenine, Cytosine, Guanine, Thymine.	Adenine, Cytosine, Guanine, Uracil.
Base Pairing	A pairs with T, C pairs with G.	A pairs with U, C pairs with G.
Chemical Stability	High stability due to deoxyribose structure.	Highly reactive and prone to degradation.
Cellular Location	Strictly inside the nucleus & mitochondria.	Synthesized in nucleus, works in cytoplasm.
Lifespan	Long-lived, permanent blueprint.	Short-lived, degraded after translation.

WHY DOES RNA FOLD?

- **Enables Enzymatic Functions:** Folding allows structural RNAs (like tRNA and rRNA) to form active, 3D catalytic pockets (ribozymes) essential for protein synthesis.
- **Maximizes Thermodynamic Stability:** Being single-stranded makes RNA chemically unstable; folding allows internal base pairing (A-U, C-G) to lower its free energy.
- **Protects Against Degradation:** Unfolded single strands are quickly destroyed by cellular enzymes (RNases); compact 3D loops and stems shield the backbone.
- **Regulates Gene Expression:** Distinct folded shapes, like riboswitches, change structure in response to cellular signals to turn translation on or off.
- **Facilitates Specific Recognition:** Unique 3D surfaces and folds allow RNA molecules to bind precisely to target proteins, lipids, or other nucleic acids.

RNA SECONDARY STRUCTURE

- refers to the specific local folding pattern of a single-stranded RNA molecule
- stabilized entirely by intramolecular hydrogen bonding between complementary bases (Adenine-Uracil and Cytosine-Guanine).
- While DNA forms a rigid, uniform double helix, single-stranded RNA loops back on itself to create diverse, dynamic 3D structural topologies essential for its cellular stability and enzymatic functions.

Common structural elements include:

1. Stem - Two complementary regions pair together.

```
AAAAUUUU
|||||||
UUUUAAAA
```

2. Hairpin Loop - One of the most common structures.

```
Stem
|||
```

Loop

AAAA

Hairpins are involved in many regulatory RNAs.

3. Internal Loop - Occurs when paired regions contain unpaired bases.
4. Bulge - One side contains extra nucleotides.
5. Multibranch Loop - Several stems meet at one junction.

WHY STRUCTURE MATTERS?

- **Dictates Function:** A linear chain cannot do work; it must fold into a 3D shape to interact chemically with the cell.
- **Creates Active Sites:** Folding brings distant segments together to form the precise "lock and key" pockets needed to catalyze reactions.
- **Ensures Specific Binding:** Unique surface geometry prevents errors by allowing a molecule to dock only with its correct partner.
- **Enables Dynamic Switching:** Molecules act as mechanical toggles, changing shapes to turn pathways or genes on and off in response to cellular signals.
- **Conserves Evolution:** Genetic sequences mutate rapidly, but the physical 3D shape remains identical across species to preserve life.
- **Drives Drug Discovery:** Knowing the exact structural coordinates allows scientists to design synthetic medicines that precisely plug and block disease targets.

WHAT IS RNAfold?

- **Core Purpose:** A fundamental command-line program in the ViennaRNA Package used to predict the secondary (2D) structure of single-stranded RNA sequences.
- **Primary Algorithm:** Uses Zuker's dynamic programming algorithm to calculate the **Minimum Free Energy (MFE)** structure - the most thermodynamically stable shape an RNA strand will fold into.
- **Input Format:** Accepts plain nucleotide sequences or standard text files in FASTA format (containing A, U, C, G bases).
- **Output Artifacts:** Generates the predicted structure in a simple text-based dot-bracket notation (where dots . are unpaired bases and brackets () are paired bases) alongside a PostScript (.ps) vector graphic diagram of the shape.
- **Advanced Metrics:** Calculates the partition function and base-pairing probability matrices to show how flexible or alternative the RNA shapes might be under physiological temperatures.

WHAT IS MINIMUM FREE ENERGY?

- **Lowest Energy State:** The most stable, resting structural state a molecule can reach when it folds .
- **Maximum Stability:** Lower values ($\Delta G < 0$) mean stronger internal bonds and a more physically favorable shape.
- **Natural Default:** Molecules spontaneously collapse into the configuration that minimizes stress and heat release.
- **Prediction Target:** Algorithms (like RNAfold) find the correct 3D shape by calculating which structure has the lowest energy score.

DOT-BRACKET NOTATION

- **Text-Based Architecture:** A clean, single-line method used to represent the 2D folded shape of an RNA sequence without drawing complex diagrams.
- **Dots (.):** Represent **unpaired** nucleotides (such as single-stranded loops, bulges, or dangling ends).
- **Opening Brackets (()):** Represent a nucleotide that is base-paired to a matching partner downstream.
- **Closing Brackets ()):** Represent the downstream nucleotide that pairs back to an opening bracket upstream.
- **Example Alignment:**
 - Sequence: G G G A A A C C C
 - Structure: (((. . .)))

(The three Gs pair directly with the three Cs to form a double-stranded stem, while the three As remain unpaired to form a hairpin loop).